

THOEN, Thailand

WMO: 483240

Lat: 17.6364N

Lon: 99.2447E

Elev: 630

StdP: 14.36

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.8	58.2	46.6	48.0	80.2	49.8	54.3	78.9	13.0	87.5	10.6	84.4	1.4	N/A	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.2	103.7	74.2	101.2	74.1	98.9	73.9	80.7	90.1	80.0	89.2	79.4	88.6	6.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	151.2	84.4	77.8	148.1	83.6	77.2	145.1	82.8	44.9	89.7	44.2	89.5	43.6	88.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	9.7	7.8	DB	49.6	106.7	3.1	2.1	47.4	108.2	45.6	109.5	43.8	110.7	41.6	112.2
			WB	48.4	82.5	3.2	0.6	46.1	82.9	44.2	83.3	42.4	83.7	40.1	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	74.3	79.1	84.4	88.2	86.0	83.4	83.0	82.5	82.5	81.0	79.1	74.4	(6)
	DBStd	5.17	4.42	3.40	3.06	4.00	3.91	2.43	2.47	1.83	2.00	2.33	3.05	4.36	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	5	4	0	0	0	0	0	0	0	0	0	0	1	(9)
	CDD50	11496	752	816	1067	1145	1115	1002	1023	1007	975	962	874	758	(10)
	CDD65	6026	291	396	602	695	650	552	558	542	525	497	424	294	(11)
	CDH74	65075	2624	4556	7936	9589	7781	5992	5930	5312	5049	4305	3587	2414	(12)
	CDH80	30852	1088	2431	4750	5815	4046	2421	2347	1924	1926	1666	1513	925	(13)
(14) Wind	WSAvg	2.2	2.1	2.7	3.2	3.1	2.7	2.1	2.4	2.0	1.7	1.5	1.6	1.8	(14)
(15) Precipitation	PrecAvg	47.6	0.4	0.5	1.2	2.5	7.4	6.0	5.6	7.3	9.6	5.9	1.1	0.4	(15)
	PrecMax	69.1	2.1	2.8	6.3	6.3	15.0	10.4	10.2	10.4	14.4	10.8	4.4	2.8	(16)
	PrecMin	33.9	0.0	0.0	0.0	0.3	1.3	1.9	3.2	4.0	5.2	1.3	0.0	0.0	(17)
	PrecStd	8.3	0.6	0.7	1.5	1.6	4.3	2.2	1.8	2.1	2.7	2.6	1.2	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.1	99.3	104.3	107.6	105.8	98.7	98.2	94.1	95.4	95.0	95.2	93.0	(19)
		MCWB	70.0	69.2	71.2	73.4	75.0	77.6	76.7	78.5	79.5	78.7	76.0	73.2	(20)
	2%	DB	91.2	97.0	101.8	104.9	101.7	94.6	94.3	92.0	93.1	92.6	92.4	90.7	(21)
		MCWB	70.2	69.8	71.7	74.2	76.7	78.3	77.0	78.3	78.9	78.2	75.7	72.8	(22)
	5%	DB	89.0	95.0	99.8	102.2	98.3	92.2	91.4	90.2	91.2	90.5	90.2	88.5	(23)
		MCWB	70.1	70.2	70.9	74.8	77.4	78.3	77.5	77.9	78.5	78.0	75.5	72.4	(24)
	10%	DB	86.6	92.5	97.6	99.5	95.3	89.9	89.3	88.4	89.1	88.3	88.2	85.6	(25)
		MCWB	70.3	70.4	71.2	75.3	77.8	78.1	77.4	77.5	78.1	77.8	75.3	71.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.5	77.0	77.7	80.1	81.5	81.1	80.2	80.9	81.5	80.8	79.8	77.1	(27)
		MCDB	85.8	89.5	93.6	94.5	92.6	89.8	88.5	89.4	91.2	89.8	89.3	88.7	(28)
	2%	WB	74.8	75.1	76.6	78.9	80.1	80.0	79.2	79.7	80.3	79.8	78.3	75.6	(29)
		MCDB	83.6	88.6	91.3	94.1	90.6	88.6	87.4	87.9	89.2	88.3	87.5	85.6	(30)
	5%	WB	73.3	73.8	75.5	78.0	79.2	79.4	78.6	78.9	79.5	79.0	77.4	74.3	(31)
		MCDB	82.1	87.2	90.5	92.9	90.2	87.8	86.7	86.9	87.8	87.4	86.1	83.5	(32)
	10%	WB	71.9	72.7	74.6	77.3	78.5	78.7	78.0	78.2	78.8	78.3	76.5	72.9	(33)
		MCDB	81.5	85.6	89.4	91.7	89.5	87.0	85.9	85.6	86.6	86.1	84.6	81.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.8	28.6	27.2	22.2	17.9	13.6	13.4	13.5	14.4	15.1	19.0	22.4	(35)
		MCDBR	28.8	32.1	31.2	26.0	22.8	17.5	16.7	16.0	17.6	18.9	22.1	25.5	(36)
	5% WB	MCWBR	9.6	8.8	7.2	5.0	4.3	4.6	4.2	4.7	5.3	6.4	7.4	8.7	(37)
		MCWBR	21.3	24.5	24.0	21.1	17.7	14.3	13.7	14.2	15.8	15.9	17.8	20.3	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.456	0.542	0.666	0.637	0.498	0.438	0.427	0.436	0.472	0.535	0.458	0.453	(40)
		taud	2.152	1.933	1.665	1.782	2.198	2.394	2.423	2.380	2.264	2.039	2.229	2.184	(41)
	Ebn at Noon	Ebn at Noon	261	244	219	226	257	271	274	273	264	243	259	258	(42)
		Edh at Noon	46	60	81	72	47	38	37	39	44	53	42	43	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1479	1687	1769	1919	1734	1507	1409	1393	1453	1456	1492	1408	(44)
	RadStd	RadStd	72	69	109	107	154	128	113	90	79	85	93	71	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (17 neighbors)	+0.47	N/A	N/A	N/A	N/A	+0.23	N/A	N/A	+218	+221	(47)

Nomenclature: See separate page